A Comparative Analysis of Asian Worker Stereotypes in US Historical Newspapers

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Abstract

This document is a supplement to the general instructions for *ACL authors. It contains instructions for using the LaTeX style file for EACL 2023. The document itself conforms to its own specifications, and is, therefore, an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Digitization of historical texts has opened up new opportunities for researchers to explore a large-scale corpus of historical texts with computational methods. Especially, the ready availability of datasets required for Natural Language Processing (NLP) research has welcomed researchers from other disciplines to NLP research (Park and Jeoung, 2022) and reduced the barriers to entry of NLP research. Taking this advantage, many researchers in diverse field, namely Sociology, History, English, and Information Science have applied NLP techniques to historical texts, such as books (Parulian et al., 2022), newspapers (Smith et al., 2013; Pedrazzini and McGillivray, 2022; Santin et al., 2016), and/or congressional records (Lin and Peng, 2022; Guldi, 2019) in addition to archival metadata (Dobreski et al., 2019). Creating an interdisciplinary research space called *Digital Humanities*, there have been studies primarily focusing on the race problem in the United States. To introduce a few large-scale computational research, Soni et al. (2021) traced the semantic change of the word, for instance when and which newspaper started to use the word with a new meaning and when and which newspapers adopted new semantic meaning of the word in abolitionist newspapers from the 19th century. Franzosi et al. (2012) analyzed racial violence, especially lynching performed by the White mob in the 19th century Georgia and presented quantitative narratives of the racial violence. These studies intersect the problem of historical racism and NLP research. However, despite the increasing attention toward the large scale text analysis on historical racism, there has been a gap of understanding the racism posed toward Asians in the United States. While it is true that the major racial tension in the United States has been between White and Black, the racial tension between White and Asian has also been a part of the history of the United States. Asian immigrant workers were derogatorily called *"coolies"*. Britannica entry on *"coolie"* introduces the word as "pejorative European usage" to describe "an unskilled labourer or porter usually in or from the Far East hired for low or subsistence wages." [1] In seeking to fill the labor shortage in the United States due to the abolition, Chinese workers were recruited and the migration of Chinese workers arrived in the United States, beyond China-neighboring countries like India and Malaysia (Farley, 1968). Perceived to be a threat to White workers by replacing them with cheap labor (Rhoads, 2002), the tension culminated in prohibiting immigration of Chinese workers (Chinese Exclusion Act of 1882) (Lee, 2002). The problem of coolie exemplifies the extension of colonial and capitalist exploitation beyond Africa and sugar-coated the extended system as indentured migrant contract workers (Van Rossum, 2016). Therefore, studying coolie problem in the United States context supplements the historical study of racial tension mostly focused between White and Black and extends the racial conflict to include Asian population. To explore understudied topic of how Asian population was discriminated in the history of the United States, we present computational text analysis on how Asian workers are represented in the U.S. newspapers. We further developed research questions as follows:

- RQ 1. How different are the semantic mean-

[1] https://www.britannica.com/money/topic/coolie-Asian-labourer

ing of "coolie" in each State?

- RQ 2. What are the words over-represented in the newspapers between outlier States and the rest?

2 Related Works

3 Methodology

3.1 Data collection

The data is collected from Chronicling America [2] API where digitized texts through optical character recognition (OCR) are accessible. When the word "coolie" was queried, API showed 124,511 pages of newspapers containing the word [3]. First, we collected the entire pages of the newspapers containing the word "coolie". Then, we extracted the text from the pages and searched the exact match for the word "coolie". This additional step ensures excluding false positive cases due to mis-recognized words. For instance, the search included the result of "cooli" even though it was not identical keyword that we wanted to query. In *New York Daily Tribune* published on August 5th, 1862 contained the word "cooli" but it accompanied many OCR errors making it doubtful whether the word "cooli" was actually from the word "coolie." [4] In order to reduce this kind of false positive cases, we double-processed the data by finding the exact match for the word "coolie" in the extracted text. This additional step resulted in 118,054 sentences containing the word "coolie" for the analysis.

3.2 Data pre-processing

We pre-processed the data by removing the punctuation, non-alphabet tokens that might have been mis-recognized during the OCR process, and stop words. The list of stop words is from NLTK package in Python and we further converted the words into lemmas using Spacy [5]. Before we feed the data into the word embedding model, we ran the FastText model (Bojanowski et al., 2017) to identify possible OCR errors. FastText embedding takes character n-gram as input and outputs the embedding vector. This model was tested effective that it can generate possible OCR error candidates (Hajiali et al., 2022). By training entire sentence that contains the word "coolie", we identified 200 most similar words to the word "coolie". For instance, "cooliez" (0.9428), "coolieb" (0.9416), and "coolieh" (0.9399) ranked high in the list of similar words to "coolie" according to the FastText embedding model. With 200 most similar words, we changed the top 200 words in the corpus into "coolie" and trained the Word2vec model in section 3.3.

3.3 RQ1. Word embedding

In order to answer RQ 1, we trained the Word2vec model (Mikolov et al., 2013) to use Continuous Bags of Words (CBOW) approach and the skip-gram approach. Both CBOW and skip-gram approach find the word embedding by predicting the target word from the context words. We trained the word2vec model with minimum word count of 5 and window size of 5 to generate the word embedding. We then took the average of the word embedding vector of the word "coolie" in each state and calculated the cosine similarity.

3.4 RQ2. Statistically over-represented words

In answering RQ 2, we use the results from the word embedding model in section 3.3. Once we identify the States that have semantically different meaning of the word "coolie", we then grouped the newspapers into two groups: the States with dissimilar semantics of the word "coolie" and the rest of the United States. We excluded sentences from the newspapers newspapers located in Virgin Islands (*n=610*) and Puerto Rico (*n=14*).

We calculated the log-odds ratio with informative Dirichlet prior (Monroe et al., 2008) by comparing the word frequency in the then-Confederacy newspapers and the newspapers published in the rest of the United States. The detailed metric is provided in equation 1.

$$\delta_w^{(i-j)} = \log \frac{y_w^i + a_w}{n^i + a_0 + y_w^i - a_w} - \log \frac{y_w^i + a_w}{n^i + a_0 - y_w^i - a_w} \quad (1)$$

The log-odds ratio with informative Dirichlet of each word w between two corpora i and j (in our study, newspapers from the then-Confederate States and the rest) given the prior frequencies are obtained from the entire corpus a. We selected 15,000 most frequent words from the entire corpus and the Z-score is calculated for each word. When

[2] https://chroniclingamerica.loc.gov/

[3] The data is collected on September 5th, 2023.

[4] https://chroniclingamerica.loc.gov/lccn/sn83030213/1862-08-05/ed-1/seq-4/ocr/

[5] https://spacy.io/

n^i is the total number of words in corpus i, y_w^i is the number of times word w appears in corpus i, a_0 is the size of the corpus a, and a_w is the frequency of word w in corpus a (Kwak et al., 2020). With the log-odds ratio, we can identify the words that are over-represented in the corpora.

4 Results

4.1 RQ1. Comparing the meaning of "coolie"

In figure 1, we present the heatmap of cosine similarity comparison across the average embedding vector of the word "coolie" in each state. We are visually informed that the cosine similarity in most of the States is high except for a few States that created a semantically different meaning of the word "coolie". For instance, Kentucky, Utah, Hawaii, and District of Columbia showed relatively low cosine similarity when their embedding vector of the word "coolie" was compared to the rest of the States. For District of Columbia, the lowest cosine similarity was found from comparing with Hawaii (0.11) while Florida showed the highest cosine similarity with District of Columbia (0.56). For Hawaii, it was Alaska (0.52) that showed the highest cosine similarity. Utah also had the semantically distant meaning of the word "coolie" when compared to Hawaii (0.21). However, the most similar State to Utah was Florida (0.53). Kentucky showed the lowest cosine similarity with Massachusetts (0.22) and Ohio was the most similar State to Kentucky (0.59). New York, Texas, and Florida also showed comparably low cosine similarity. The cosine similarity between New York and Hawaii was 0.27 and Wisconsin and Wyoming were the most similar States to New York (0.55). Texas also had the most dissimilar meaning of "coolie" when compared to New York (0.35) and the most similar State to Texas was Ohio (0.63). For Florida, the most dissimilar State was Hawaii (0.37) and Connecticut was the most similar State to Florida (0.67).

To delve into the semantic difference of the word "coolie" in each State, we present the top 10 most similar words to the word "coolie" in each State.

4.2 RQ 2. Statistically over-represented words in the South and the rest of the United States

4.3 Discussion

5 Limitations

6 Future Work

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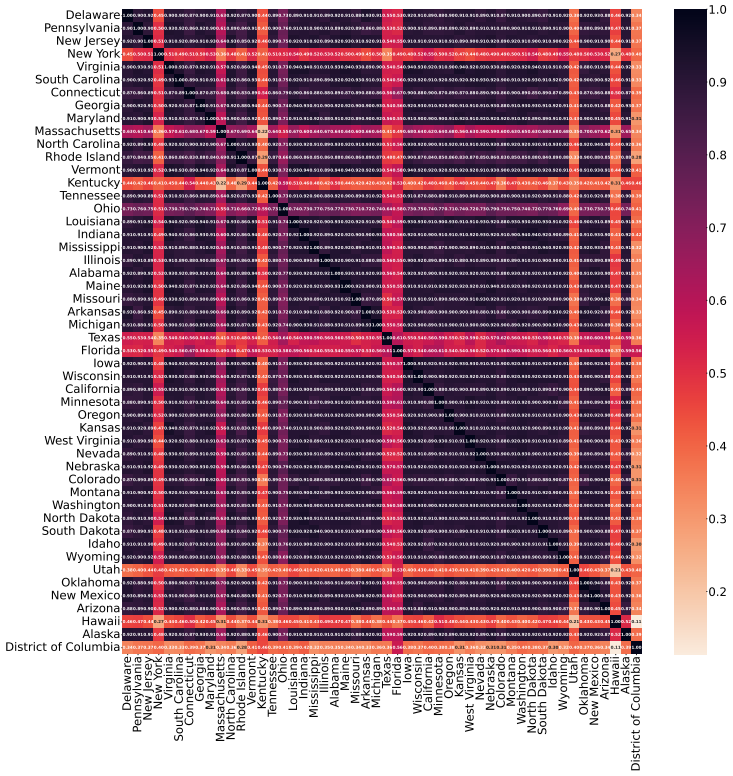

Figure 1: The heatmap of cosine similarity comparison across the average embedding vector of the word “coolie” in each state

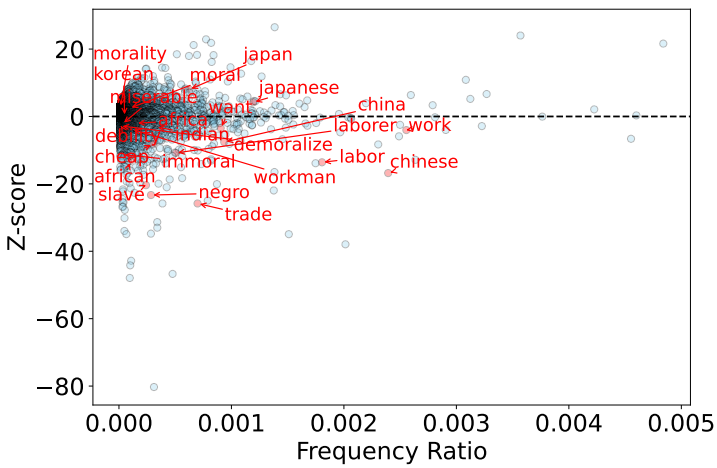

Figure 2: The Z-score of words in the newspapers

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A Example Appendix